

Orientation

Three Essays on the Scarcity of Navigation
in Minds, Institutions, and Knowledge Traditions

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Preface

These three essays share a single observation: production scales more easily than orientation.

That claim sounds modest. Its consequences are not.

By production is meant any process that generates outputs: candidate solutions, institutional constraints, validated results. Production is the visible, measurable dimension of any knowledge-creating system. It is what gets counted, rewarded, and accelerated. By orientation is meant the activity that keeps production aimed — that maintains coherent objectives through time, preserves the reasons behind inherited structures, and constructs integrative understanding across accumulated results. Orientation is the invisible, deferred, and systematically underrewarded dimension of the same systems.

When the two scale at different rates, something accumulates. This series names three forms of that accumulation.

Scale	Production	Orientation	Accumulation
Individual	Search	Navigation	Navigability debt
Institution	Constraint accumulation	Reason reconstruction	Reason decay
Knowledge tradition	Knowledge production	Synthesis	Understanding debt

The first essay, *Orientation*, begins at the individual scale. It observes that recent systems capable of generating candidate solutions at high speed have broken a coupling that previously constrained inquiry: the coupling between exploration and implementation. When generation was expensive, exploration was bounded by it. You could not examine more possibilities than you could afford to develop. That constraint is now substantially relaxed. The space of reachable candidates has expanded dramatically.

What has not expanded is the capacity to navigate that space while remaining coherently aimed. Search and orientation are not symmetric. Search scales readily when generation becomes cheap. Orientation — the capacity to maintain coherent objectives through time, to distinguish progress from drift, to evaluate trajectories rather than states — remains tied to cognitive resources that do not scale with generation capacity. The bottleneck shifts. Generation was the limiting resource. Now orientation is.

The first essay develops the consequences of this shift: why exploration must precede formalization rather than follow from specification, why goal continuity requires deliberate mechanisms that look like inefficiency from inside a search-maximizing frame, why repair preserves navigability rather than merely correcting errors, and what it means formally for a goal to undergo projection collapse — to be optimized against a shadow of itself across many locally reasonable iterations.

The second essay, *Orientation and Institutions*, moves to the institutional scale. Here the relevant production process is not the generation of candidate solutions but the accumulation of constraints: the rules, procedures, precedents, and commitments through which an institution encodes its accumulated experience. Constraints are durable. They can be copied, transmitted, and enforced by people who have never encountered the situations that made them appropriate. What cannot be copied is the argument that made each constraint intelligible — the understanding of why this constraint, in this domain, in response to these failures, was the right one.

Reasons decay faster than constraints. Every generation can inherit the rule by transcription. Every generation must reconstruct the reason through something much closer to apprenticeship than copying. When the transmission of reasons is compressed — when apprenticeship is shortened, mentorship reduced, case reasoning replaced by procedure lookup — the institution retains its formal shape while losing its orientation. Constraints that were originally reasoned become ritualized: followed but no longer understood. Ritualized constraints produce institutional brittleness that is invisible in stable conditions and catastrophic in novel ones.

The second essay introduces the Reason Retention Principle: an institution remains oriented only to the extent that it can reconstruct the arguments that originally justified its constraints. Reconstruction is not archival retention, which preserves words, and it is not recitation, which preserves formulas. It is the capacity to regenerate the original argument from first principles and apply it to situations the original designers did not anticipate. When that capacity is lost, the institution enters a self-sealing condition: every internal metric is consistent with current operative objectives, and the mechanisms required to detect divergence from founding objectives have themselves degraded.

The third essay, *Orientation and Knowledge*, extends the argument to civiliza-

tional scale. Knowledge traditions face a structural analog of both prior failures, but operating across longer time horizons and larger communities than any institution. The relevant production process is knowledge production itself: the generation of validated results, proofs, datasets, and techniques. Knowledge accumulates monotonically. Results are added to the literature and rarely removed. Understanding does not accumulate in the same way.

Understanding is the construction of coherent structure across knowledge. It cannot be stored or transmitted by copying; it must be reconstructed by each new understander, at increasing cost as the body of knowledge grows and fragments. When knowledge production exceeds the rate at which coherent structure is built across results, understanding debt accumulates. Understanding debt is not a simple backlog. It compounds: deferred integration shapes subsequent knowledge production, which occurs in the absence of the integrations that would have constrained it, deepening the fragmentation that synthesis must eventually overcome.

The third essay introduces two further distinctions that apply retroactively to the first two. A causal trajectory is the sequence of decisions that produced a current state. A selection trajectory is the set of alternatives that were rejected and the criteria by which they were rejected. Selection trajectories are harder to preserve than causal ones, because rejected alternatives leave weaker traces than surviving paths. The literature stores survivors. Understanding stores the selection history — the record of what was tried, what failed, and what the surviving formulations ruled out. When the selection history disappears, survivors lose the context that made them significant. Results can be cited but not built upon in the deepest sense. Constraints can be followed but not adapted to novel situations. Goals can be pursued but not revised when they should be.

The three essays are independent. Each can be read without the others. But they are designed as a series, and the series has a shape.

The first essay identifies an asymmetry at the smallest scale examined — the individual mind — and derives a set of consequences about how inquiry should be structured when generation is abundant. The second essay shows that the same asymmetry appears at institutional scale, where it produces a failure mode that is both older and slower than the individual version: not projection collapse across iterations but institutional drift across generations. The third essay shows

that the asymmetry appears again at civilizational scale, where it produces fragmentation across the body of what a tradition has learned.

Each scale introduces something new. The first introduces the search–orientation asymmetry and the concept of navigability debt. The second introduces the reason–constraint asymmetry and the phenomenon of ritualization — the intermediate state between reasoned constraint and abandoned constraint that is probably the most common condition of inherited institutional rules. The third introduces the knowledge–understanding asymmetry, the distinction between causal and selection trajectories, and the compounding dynamics of understanding debt.

The formal vocabulary develops across the series. Projection collapse, introduced in the first essay to describe individual goal drift, reappears in the second as the formal model of institutional drift and in the third as the model of founding-question divergence across subfields. The three-component context decomposition introduced in the second essay — current state, archived memory, and reconstructive capacity — maps onto the knowledge graph structure of the third. The debt integral introduced in the third applies retroactively to the navigability and reason-decay accumulations of the first two.

The principle that unifies all three is the Trajectory Preservation Principle, stated in the third essay:

A system remains oriented only to the extent that future agents can reconstruct the trajectories — both causal and selective — that generated its present state. Orientation failure occurs when state accumulation exceeds trajectory reconstructability.

Navigability debt, reason decay, and understanding debt are three manifestations of this single process at three scales. In each case, a productive process generates states. An orientation process maintains access to the trajectories that produced those states. When the productive process accelerates and the orientation process does not keep pace, the trajectories become inaccessible. The states remain. They are not wrong. They are simply no longer connected to the process that produced them, and that disconnection is what disorientation means.

There is a simpler formulation that runs alongside the formal one. Every debt in this series arises because systems preserve nouns more effectively than verbs. Search results more effectively than navigation. Constraints more effectively

than reasons. Knowledge more effectively than understanding. States more effectively than trajectories. In each case, the residue of the process is preserved while the process that produced the residue is lost. Civilization becomes dis-oriented when it preserves the products of processes more effectively than the processes themselves.

What accelerates is always visible. What preserves orientation is always invisible until it is gone.

<i>Essay</i>	<i>Central Principle</i>
Orientation	As search becomes abundant, orientation becomes scarce.
Orientation and Institutions	An institution remains oriented only to the extent that it can reconstruct the arguments that originally justified its constraints.
Orientation and Knowledge	The literature stores survivors. Understanding stores the selection history.

Orientation

On the Scarcity of Navigation in an Age of Abundant Search

Abstract

For most of the history of knowledge work, exploration and generation were tightly coupled: to investigate a possibility one had to implement it, and implementation was costly. Recent developments in automated generation have broken this coupling. The cost of producing candidate solutions has collapsed dramatically, expanding the range of possibilities that a single agent can examine in a given interval of time. This essay argues that the primary consequence of this change is not an increase in productive capacity but a shift in which resource is limiting. When generation was expensive, the bottleneck was search. When generation becomes abundant, the bottleneck shifts to orientation — the capacity to maintain a coherent objective through time, to distinguish progress from drift, and to determine when exploration should give way to commitment. Orientation does not scale with search. The mechanisms that preserve goal continuity, support iterative repair, and sustain the distinction between exploration and formalization become more valuable rather than less as search capacity increases. The central problem of an age of abundant search is no longer how to generate possibilities but how to remain oriented among them.

1. Search Becomes Abundant

Something genuinely unprecedented has happened to the economics of exploration.

For centuries, the cost of examining an idea was proportional to the cost of developing it. A scientist wishing to compare three competing hypotheses had to run three experiments. An architect wishing to evaluate alternative structural

systems had to produce three sets of drawings. A programmer wishing to explore three approaches to a design problem had to implement three programs. Exploration was bounded by implementation capacity, and implementation capacity was scarce. The practical consequence was that inquiry was forced to narrow before it widened. Exploration had to be preceded by sufficient prior reasoning to justify the expenditure of implementation effort. The thinker compressed the search space mentally before exploring it empirically.

Recent systems capable of generating candidate solutions at high speed and low marginal cost have substantially altered this relationship. The collapse of generation costs means that a single individual can now explore regions of possibility space that would previously have required teams of specialists or months of concentrated labor. Three candidate architectures can be sketched in hours rather than weeks. Ten alternative formulations of a problem can be examined before committing to any one of them. The cost that previously compelled premature convergence has been dramatically reduced.

This is a genuine enlargement of human cognitive capacity. It is worth dwelling on the magnitude of the change before examining its consequences. The ability to defer commitment, to explore widely before narrowing, to hold multiple candidate structures in play simultaneously — these are not trivial affordances. Scientific method in practice has always been constrained by the cost of generating and testing hypotheses. Design practice has always been constrained by the cost of prototyping. The relaxation of those constraints, even partially, represents a qualitative shift in how inquiry can proceed.

The question is what this shift implies for the overall structure of productive work.

A common answer is that increased generation capacity translates directly into increased productive output. If generation is faster, more can be produced in a given time. This answer is not wrong, but it is incomplete in a way that matters. It focuses on what the expansion of search capacity makes easier while neglecting what it leaves unchanged — and what it makes harder.

2. The Asymmetry Between Search and Orientation

To identify what the expansion of search leaves unchanged, it is useful to distinguish two components of any goal-directed inquiry.

The first component is *search*: the process of generating candidate states, solutions, hypotheses, or structures. Search explores a possibility space. It asks, in various ways, what could exist here. The outputs of search are candidates — proposals that may or may not satisfy the objectives of the inquiry.

The second component is *orientation*: the process of maintaining a coherent objective through time, evaluating candidates against that objective, determining which regions of possibility space are worth exploring, and recognizing when the inquiry has made genuine progress as opposed to local movement. Orientation asks, in various ways, what is worth reaching and whether one is still moving toward it.

These two components are not symmetric in how they respond to changes in generation capacity.

Search scales readily. When generation is cheap, more candidates can be produced. The range of positions, architectures, solutions, and alternatives available for examination expands. The breadth of exploration increases. More of possibility space becomes accessible per unit of time.

Orientation does not scale through the same mechanisms as search. It is worth being precise about why, because orientation is not a single capacity. It comprises at least four distinguishable functions. The first is *evaluation*: determining whether a candidate satisfies some criterion. The second is *goal maintenance*: keeping the original objective in view across an extended sequence of local decisions. The third is *trajectory assessment*: judging whether the current path, not merely the current state, is moving toward the objective. The fourth is *goal revision*: recognizing when the original objective was itself mistaken or incomplete, and adjusting it without losing continuity with what was genuinely intended.

These four functions scale differently. Evaluation, when the criterion is well-specified and computable, can often be automated or parallelized. It is the component of orientation most amenable to computational assistance. Goal maintenance is harder to delegate, because it requires a representation of the objective that is not itself a product of the local context. Trajectory assessment is harder still, because it requires comparing sequences rather than states. Goal revision is the least tractable of all, because it requires the kind of value-laden judgment that cannot be reduced to optimization against a fixed criterion — it is, in a sense, the meta-orientation that governs all the others.

The characteristic error of an age of abundant search is to treat evaluation as though it were the whole of orientation, and to conclude that because evaluation can scale, orientation can scale. The higher levels of orientation — goal maintenance, trajectory assessment, and especially goal revision — remain substantially tied to the cognitive and institutional resources of the inquiring agent. They require continuity of judgment that cannot be easily parallelized or offloaded.

Principle I.1. *Search and orientation scale at different rates. Search scales computationally, through parallelism and faster generation. Orientation scales institutionally and cognitively, through review, shared constraints, memory systems, and the continuity of judgment. Confusing these scaling regimes is the characteristic error of an age of abundant search.*

Principle I.2. *As search becomes abundant, orientation becomes scarce.*

A Concrete Example

The distinction between search and orientation is not unique to automated generation systems. Similar transitions appear whenever technological advances expand the set of reachable states faster than the capacity to evaluate them.

A recent example comes from dense biomedical image registration. Jena, Chaudhari, and Gee describe a registration framework that dramatically reduces the computational cost of aligning large volumetric datasets across modalities and species [1]. Hyperparameter studies that would previously have required years of computation become feasible in hours, enabling researchers to explore hundreds of candidate configurations before committing to any single registration strategy.

The significance of this result is not merely speed. The enlarged computational budget expands the region of parameter space that can be examined before commitment. More candidate solutions become reachable. Yet the scientific problem remains unchanged. Researchers must still determine which metrics matter, which regions of parameter space correspond to biologically meaningful solutions, and when exploration has been sufficient to justify formalization. The search space expands. The requirement for orientation does not. Indeed, the need for orientation becomes more acute precisely because more possibilities can now be examined without the natural constraint that expense formerly

imposed.

The practical implication of this asymmetry follows directly. When both search and orientation were expensive, the limiting resource was search. Productive work was constrained primarily by the ability to generate candidates. But as search becomes abundant, the asymmetry becomes visible. Orientation, which was never the bottleneck when search was expensive, becomes the bottleneck when search becomes cheap. The limiting factor in productive inquiry shifts from the capacity to explore to the capacity to remain oriented within an increasingly large explored space.

This shift has not been widely recognized, for a simple reason: the expansion of search capacity is highly visible, while the constraint imposed by orientation is largely invisible. One can count candidates generated. One cannot easily count orientations maintained. The measurable thing improves dramatically; the unmeasured thing remains constant; and the temptation is to interpret the improvement in the measurable thing as an overall increase in productive capacity. But what has increased is search. What has not increased is the capacity to make use of search while remaining coherently aimed.

3. Exploration Before Formalization

The asymmetry between search and orientation has direct implications for how productive inquiry should be structured.

A common model of goal-directed work runs as follows. The agent specifies the objective in advance. The specification is then handed to a process — human or automated — that implements it. The output is evaluated and refined. This model has the virtue of clarity. The objective is explicit. The implementation follows from the specification.

The difficulty with this model is that it assumes the agent can produce a complete and accurate specification before exploration has occurred. In practice, specifications are hypotheses about what is wanted. They may be partially correct. They will almost certainly be incomplete. They will contain assumptions that appear reasonable before exploration but reveal themselves as mistaken once alternatives are examined. The specification encodes the agent's prior beliefs about the solution space. Prior beliefs about solution spaces are often wrong in ways that can only be discovered by exploring the space.

Productive inquiry, whether in science, design, engineering, or any other domain requiring the navigation of a large possibility space, characteristically follows a different structure. Exploration precedes formalization. The agent examines the territory before committing to an architecture. Prototypes are built before interfaces are specified. Experiments are run before theories are committed to paper. The laboratory prototype is not a defective version of the final instrument. It is a device for exposing structure that the inquirer does not yet have access to. Once the structure has been exposed, formalization can proceed from genuine understanding rather than from prior belief.

Observation I.1. *Formalization is an irreversible operation. Every abstraction eliminates possibilities. Every interface commitment closes off alternative structures. Every architectural decision forecloses some future modifications. The cost of premature formalization is therefore not merely that the resulting structure is suboptimal, but that it is difficult to revise. Premature commitment is not an implementation error but an orientation error: it eliminates regions of possibility space before their value has been established.*

The expansion of search capacity makes the separation between exploration and formalization more important, not less. When generation is cheap, the quantity of candidates that can be produced before commitment is dramatically larger. The territory that can be explored before formalization is much wider. But exploiting this expanded exploration capacity requires maintaining the distinction. If cheap generation is used to implement specifications more quickly rather than to explore more widely before committing to specifications, the new capacity is being used to do the old thing faster rather than to do something qualitatively different. The result is more rapid convergence on a solution that may be no better chosen than before — merely more quickly arrived at.

The inquiry that benefits most from abundant search is inquiry that defers formalization until exploration has been sufficient to justify commitment. The exploration phase exposes structure. The formalization phase stabilizes it. Collapsing them eliminates the epistemic benefit of cheap generation.

The biomedical registration case is instructive here as well. The authors emphasize that optimal registration parameters are not universal: they differ across modalities, species, and experimental objectives [1]. Faster computation does not eliminate the need to determine which parameters are appropriate for which

contexts. It makes the exploration of that question tractable for the first time. The system does not assume the correct parameters in advance and implement them efficiently. It expands the space that can be explored before any commitment is made. That is precisely the structure this section is describing: abundant search enabling genuine exploration, rather than more rapid convergence on a prior assumption.

4. Goal Continuity

A further consequence of the asymmetry between search and orientation concerns what might be called goal continuity: the preservation of the agent's original objective through an extended sequence of local decisions.

In any iterative process, each step produces an intermediate state. The agent evaluates the intermediate state, makes a decision about the next step, and proceeds. In a short process, with few steps, the original objective remains clearly in view throughout. But in a long iterative process, the relationship between each local step and the original objective can become increasingly indirect. The agent is evaluating the intermediate state, not the objective directly. The decision about the next step is made relative to the intermediate state. If the intermediate state is an imperfect proxy for the objective — if it captures some aspects of what was originally wanted while losing others — then each successive step may optimize the proxy rather than the objective.

This is not a new problem. It is recognizable in any long-horizon planning process, in any multi-stage optimization, in any collaborative project where the original intent must be transmitted across many hands over extended time. What changes when search becomes abundant is not the structure of the problem but its severity. When search is expensive, iterative processes are necessarily short. The number of steps between the original specification and the final output is limited by the cost of each step. There is less opportunity for the objective to be progressively displaced by its proxy.

When search becomes cheap and iteration becomes rapid, the number of steps that can occur between the original specification and any given intermediate state grows dramatically. The original objective must now persist through far more iterations. The accumulated displacement across many cheap steps can exceed the displacement that would have occurred across a few expensive ones.

Definition I.1. *Let an objective G be given at the outset of an inquiry. At each step t , the agent's operative goal is a projection G_t of G onto the locally visible context. A projection discards some aspects of G while preserving others. When the context is impoverished relative to G — when only a small portion of the original objective is visible from the current intermediate state — the projection G_t may differ substantially from G while appearing locally coherent. Over a sequence of steps, the cumulative displacement $\|G - G_t\|$ may grow even when each individual transition $\|G_{t-1} - G_t\|$ is small.*

The failure mode this describes is distinctive. No individual step is obviously wrong. Each local decision may appear reasonable given what is currently visible. The trajectory as a whole, however, diverges from the original aim. The system optimizes a shadow of the goal rather than the goal itself.

A simple illustration makes the structure concrete. Suppose the original objective is to build a research platform that is flexible enough to accommodate unanticipated future requirements. At each iteration, the locally visible problem differs: in the first, the system is too slow; in the second, it is too complex to explain to collaborators; in the third, it scores poorly on a standard benchmark; in the fourth, it is difficult to deploy. Each optimization is reasonable. Faster is better; simpler explanations are better; benchmark scores matter; deployment ease matters. Yet the cumulative effect of optimizing for speed, then simplicity, then benchmark score, then deployment ease is a system that is specialized, well-documented for its current use, competitive on current metrics, and easy to deploy in its current configuration — but no longer flexible. The original goal has disappeared not through any single bad decision but through the accumulated displacement of many locally good ones. The context visible at each step included the current problem and excluded the original objective. Each projection was impoverished in the same direction.

Mechanisms that preserve goal continuity are therefore not merely useful supplements to rapid generation. They are the structural counterpart without which cheap search produces trajectories rather than progress. Such mechanisms include: explicit review that compares intermediate states against original objectives rather than against the immediately preceding state; deliberate re-articulation of the objective at intervals throughout the inquiry; maintenance of documentation that preserves the reasoning behind early commitments; and periodic stepping back from local improvement to assess whether the overall

trajectory remains coherent.

None of these mechanisms are glamorous. None of them resemble the visible productivity gains associated with rapid generation. But their value increases precisely as search becomes cheaper. When generation is abundant and iteration is rapid, the risk of cumulative displacement increases. The mechanisms that counteract displacement become proportionally more important, not less.

5. Repair as Navigation

The preceding sections have described several ways in which orientation, goal continuity, and the structure of productive inquiry change when search becomes abundant. A common thread connects them: activities that appear to be about maintaining or recovering from failure are actually about preserving navigability through time.

The dominant view of repair in complex systems treats it as cleanup. Something has gone wrong; repair restores the system to correct functioning; inquiry continues. On this view, a well-designed process would minimize the need for repair. Repair is a cost, and its reduction is a goal.

An alternative view treats repair as the primary mechanism by which living systems — systems that persist and adapt through time rather than being built once and completed — maintain the capacity for future movement. On this view, repair is not evidence of failure. It is the activity through which a system stays navigable. A system that is never repaired is a system whose trajectory has been fixed. A system that is regularly repaired is a system that continues to have a trajectory.

Principle I.3. *Repair is not the correction of deviation from a fixed target. It is the process by which an adaptive system maintains the conditions for continued navigation. The value of repair is not that it restores a previous state but that it preserves the possibility of future states.*

The distinction between these two views has practical consequences that become clearer as search capacity increases. When generation is cheap and output is abundant, a natural response is to regard repair as increasingly unnecessary. If producing more output is nearly free, why invest in maintaining what has already been produced? The cost-benefit calculation appears to favor generation

over repair.

This calculation neglects a dimension that does not appear in the immediate cost structure: navigability. A system that has been extended without repair accumulates structural commitments that were made in the context of earlier states. Those commitments may no longer be appropriate given later developments. They may be locally invisible — no individual component of the system is obviously wrong — but they constrain the range of future modifications. The cost of those constraints does not appear until future modifications are attempted. At that point, the accumulated navigability debt becomes apparent.

The objection to rapidly generated, unrepaired systems is therefore not that any given component is incorrect. It is that the future becomes difficult. The system enters states from which many desirable future states are unreachable. This is a reachability failure rather than a correctness failure. The system is not wrong in the sense of failing to match a specification. It is wrong in the sense of having closed off paths that might have led somewhere valuable.

Repair, in this light, is navigation. It is the activity through which the agent maintains access to the future. Review, refactoring, constraint formation, interface redesign, and explicit articulation of module boundaries are all forms of repair. They appear inefficient from the perspective of immediate generation. They are indispensable from the perspective of continued navigability.

The value of these activities is not constant. It increases as the rate of generation increases. A system generated slowly accumulates navigability debt slowly. A system generated rapidly accumulates navigability debt rapidly. The faster the search, the more important the repair.

An instructive analogy appears in diffeomorphic image registration. The objective of such a procedure is not merely to move one image into correspondence with another. The transformation must preserve topology throughout the process: no tearing, no folding, no destruction of the structural relationships that make the final alignment meaningful [1]. A registration that reaches the correct endpoint while introducing topological defects is regarded as defective even if the final alignment appears locally successful. The reason is that topology preservation constrains the admissible trajectory rather than merely the endpoint. Quality depends not only on where the transformation arrives but on what structural properties remain intact after arrival.

Repair in cognitive and technical systems plays an analogous role. Its purpose

is not to optimize the current state but to preserve the structural conditions that keep future states reachable. A system that is locally correct but topologically compromised — one whose accumulated navigability debt has closed off paths that cannot easily be reopened — has failed in the same sense as a registration that achieves correspondence while destroying the structure it was meant to preserve.

6. The New Bottleneck

The preceding analysis can be summarized as a sequence of displacements.

When search was expensive, the bottleneck in productive inquiry was the capacity to generate candidates. More thinking, more design effort, more implementation capacity — these were the resources whose expansion would accelerate inquiry most directly. Orientation was also required, but it was rarely the limiting factor, because the cost of search naturally constrained the length of any iterative process and thereby limited the opportunity for cumulative displacement.

When search becomes cheap, the bottleneck shifts. Generation is no longer limiting. The capacity to orient — to maintain coherent objectives through extended iterations, to distinguish exploration from formalization, to preserve goal continuity against cumulative displacement, to repair for navigability rather than for immediate correctness — becomes the resource whose scarcity most constrains productive inquiry.

This shift has several implications.

First, activities that increase search capacity without attending to orientation may produce less than their apparent output suggests. The visible quantity of generated material increases. The proportion of that material that is genuinely useful — that has been selected, evaluated, and integrated in relation to a well-maintained objective — may not increase proportionally and may in some cases decrease.

Second, the contribution of human judgment to any collaborative inquiry between a person and a generation system is not symmetric with the system's contribution. The system contributes search. The human contributes orientation. These are not interchangeable. A more powerful search system does not reduce the need for orientation; it increases it, because a larger explored space requires

more careful navigation. The appropriate response to more capable generation is not less human involvement in the inquiry but differently structured human involvement — concentrated in the orientation activities that search cannot perform.

Third, the skills most valuable in productive inquiry have shifted. When search was expensive, implementation skill — the ability to generate high-quality outputs efficiently — was central. When search becomes abundant, implementation skill remains necessary but is no longer distinctive. What becomes distinctive is the capacity for orientation: the ability to maintain coherent objectives through extended iteration, to recognize when exploration should give way to commitment, to evaluate trajectories rather than states, and to repair for navigability rather than for immediate correctness. These are not new skills. They are skills that good inquiry has always required. What has changed is their relative value.

Fourth, the appropriate structure of productive work changes. When generation was expensive, it made sense to minimize the number of iterations by investing heavily in specification before implementation. When generation is cheap, this logic reverses: it becomes sensible to invest less in prior specification and more in exploration, using abundant generation to examine the territory before committing to an architecture. But exploiting this reversal requires maintaining the discipline of orientation throughout the exploration phase, so that the insights gained from exploration are not lost when formalization eventually occurs.

7. Formal Interpretation

The preceding account has been developed in ordinary language, with the aim of making the central asymmetry accessible without presupposing any particular formal apparatus. It may nevertheless be useful to indicate briefly how the observations can be expressed in a more precise vocabulary.

The concept that provides the most direct formal grip on the phenomena described here is *projection collapse*.

Let G denote the goal defined at the outset of an inquiry. At each stage t of an iterative process, the agent does not have direct access to G . The agent has access to the current intermediate state S_t and to some context C_t capturing what is locally visible from that state. The operative goal at stage t is a projection $\pi(G, C_t)$: the image of G restricted to the locally visible context. Projection is

necessarily lossy. When C_t is rich, $\pi(G, C_t)$ is faithful and optimization serves G well. When C_t is impoverished, the projection is distorted and optimization relative to $\pi(G, C_t)$ may diverge substantially from G .

Definition I.2 (Projection Collapse). *A sequence of projections $\{\pi(G, C_t)\}_{t=0}^T$ undergoes projection collapse if the cumulative displacement $d(G, \pi(G, C_T))$ grows substantially over the sequence, even when each individual step displacement $d(\pi(G, C_{t-1}), \pi(G, C_t))$ is small. Projection collapse occurs when iterative local optimization produces global divergence from the original objective.*

Projection collapse is the formal analog of cumulative goal displacement. The trajectory appears locally coherent: each step is reasonable given what is currently visible. The sequence as a whole, however, moves away from the original aim, because each step optimizes a projection that has already lost some information about G .

The phenomena described in earlier sections map directly onto this structure. Exploration before formalization is the practice of ensuring C_t remains rich before commitment narrows the accessible context. Goal continuity mechanisms — review, re-articulation, documentation of early reasoning — are practices for maintaining a representation of G that is not entirely determined by C_t , periodically re-anchoring the operative goal against accumulated drift. Repair as navigation is the practice of preventing structural commitments from impoverishing the context to the point where future states become unreachable.

Orientation, in these terms, is the capacity to maintain a faithful correspondence between $\pi(G, C_t)$ and G across an extended sequence — to detect when the projection has become impoverished, and to act on that detection. This capacity cannot be delegated to any process that operates only within the local context. It is the irreducible contribution of the inquirer to any inquiry in which generation has been substantially externalized. As search scales, that contribution does not diminish. It becomes the limiting resource.

Conclusion

The collapse of generation costs has expanded search capacity in ways that are genuine and substantial. More of possibility space is accessible to individual inquiry than at any previous point. More alternatives can be examined before commitment. More candidate structures can be held simultaneously in play.

These are real enlargements of what inquiry can accomplish.

What has not expanded is orientation capacity. The higher levels of orientation — goal maintenance, trajectory assessment, and goal revision — remain tied to the cognitive and institutional resources of the inquiring agent. They do not scale computationally. They become relatively more scarce as search becomes relatively more abundant.

The consequence is a shift in what matters most. When generation was the bottleneck, generating well was the primary challenge. When generation becomes abundant, remaining oriented becomes the primary challenge. Activities that appear to be about maintenance, review, repair, and explicit goal articulation are not inefficiencies to be minimized. They are the mechanisms through which productive inquiry remains productive as the scale of search increases.

This principle does not depend on any particular technology or moment. It applies wherever the capacity to generate candidates substantially exceeds the capacity to evaluate trajectories. It is as relevant to scientific programs that have grown faster than their conceptual frameworks, to institutions that have expanded faster than their governance structures, and to bodies of knowledge that have accumulated faster than their organizing principles, as it is to any other domain where search and orientation have become uncoupled.

The central problem of an age of abundant search is not how to generate possibilities. It is how to navigate among them without losing the goal that made them worth exploring.

Orientation and Institutions

How Organizations Preserve Goals Across Time

Abstract

Institutions routinely outlive the reasons for their own rules. The familiar explanations for this phenomenon locate the failure in agents: in capture, incompetence, bad incentives, or the self-interest of those who administer the rules. This essay argues that those explanations are incomplete because they treat institutional drift as a contingent failure rather than a structural tendency. The structural explanation is that constraints persist more readily than the reasons that justified them. When reasons decay while constraints remain, the institution retains its formal shape while losing its orientation. This essay calls that process institutional projection collapse. The central claim is a Reason Retention Principle: an institution remains oriented only to the extent that it can reconstruct the arguments that originally justified its constraints. Reconstruction, not mere archival preservation, is the operative requirement. A second structural claim follows: projection collapse becomes increasingly difficult to detect as the institution loses access to the reasons that generated its present constraints. Sufficiently advanced drift destroys the mechanisms required to recognize drift. These two claims together reframe a large class of institutional failures as orientation failures — problems of continuity rather than problems of incentive.

1. Constraints Without Reasons

Institutional drift is a familiar observation. Organizations formed around a clear purpose accumulate procedures, departments, metrics, and requirements until the original purpose becomes difficult to locate inside the structure that was built to serve it. Universities formed to pursue knowledge optimize publication counts. Regulatory agencies formed to protect the public expand procedu-

ral compliance. Professional bodies formed to maintain standards develop elaborate certification requirements disconnected from the competences those requirements were meant to ensure. Companies formed to solve a concrete problem maintain internal reporting structures long after the problem has changed or disappeared.

The standard explanations for this pattern appeal to agents. Capture: those who administer rules develop interests in perpetuating them. Bureaucracy: formal structures generate their own momentum independent of the goals they ostensibly serve. Bad incentives: the metrics available for measuring institutional performance diverge from the objectives the institution was formed to pursue. Agency problems: the people who run institutions are not identical with the people the institutions were formed to benefit, and their interests differ. These explanations are not wrong. Capture happens. Bureaucracy is real. Incentive misalignment is common.

But these explanations are incomplete in a way that matters. They treat institutional drift as contingent — as something that happens when agents behave badly or when incentive structures are poorly designed. They imply that better agents, better incentives, or better monitoring would prevent the failure. What they do not explain is why institutional drift is so universal, so persistent, and so difficult to reverse even in institutions that begin with good agents, good incentives, and serious monitoring. They do not explain why the failure so often occurs invisibly, without any point at which someone decided to abandon the original objective.

The structural explanation begins differently. Constraints are durable. Reasons are fragile. A rule can be copied into a new document, transmitted to a new generation of administrators, and enforced by people who have never encountered the situation the rule was designed to address. What cannot be copied by transcription alone is the argument that made the rule intelligible — the understanding of why this constraint, in this domain, was necessary. That understanding requires not only the words of the original justification but the background knowledge, practical experience, and interpretive capacity required to make those words meaningful.

Institutional drift, on this account, is not primarily a failure of agents. It is a failure of transmission. What decays is not the constraint but its reason. And once the reason is gone, the constraint is available for a new purpose:

not the purpose it was designed to serve, but whatever purpose the current administrators find locally convenient to attribute to it.

The asymmetry in transmission cost explains why this failure is structural rather than contingent. Copying a constraint costs approximately nothing. Every generation can inherit the rule by transcription. Transmitting the reason for a constraint costs far more: it requires extended engagement, a recipient capable of understanding the original problem, and a context rich enough to make the reasoning legible. Each generation must re-earn the reason through something closer to reconstruction than inheritance. Constraint transmission is mechanical. Reason transmission is social. Under any pressure that reduces the time and attention available for the social work of transmission, reasons decay while constraints persist. The institution retains its shape while losing its orientation.

2. Search Velocity and Contextual Depth

The previous essay in this pair established that search and orientation scale at different rates. When the cost of generating candidates falls, the set of reachable states expands, but the capacity to evaluate trajectories does not expand proportionally. The bottleneck shifts from production to navigation.

The same asymmetry appears at institutional scale, but its form is different. The relevant scarcity is not between generating options and choosing among them. It is between the accumulation of institutional decisions and the preservation of the context that made those decisions intelligible. Institutions generate constraints continuously — through policies, precedents, procedures, commitments, and agreements. What they accumulate more slowly, and lose more easily, is the body of reasons that made each constraint appropriate in its original context.

It is useful to characterize different institutional forms along a single axis: the ratio of search velocity to contextual depth. Search velocity measures how rapidly an institution can produce and implement new actions, decisions, and policies. Contextual depth measures how much accumulated reasoning must be consulted before an action is considered legitimate.

Principle II.1 (Contextual Depth). *Institutional responsiveness increases as required contextual depth decreases, but orientation becomes increasingly vulnerable to projection collapse. Institutions differ not only in efficiency but in how much of their accumulated*

orientation must be engaged before legitimate action can be taken.

High search velocity with low contextual depth enables rapid adaptation but provides little protection against projection collapse. Each new decision is made in a thin context. Early constraints are quickly overwritten rather than built upon. The institution remains agile but its trajectory is easily deflected by local conditions. Low search velocity with high contextual depth provides strong protection against projection collapse but may prevent the institution from responding to genuine changes in its situation. Each new decision must negotiate a dense accumulated context that may itself have drifted.

This is a pressure, not a law. Institutions can resist it. But resisting it requires deliberate investment in exactly the mechanisms that search-velocity pressure tends to erode: the maintenance of interpretive communities who can engage the accumulated context, the preservation of reasons alongside rules, and the cultivation of trajectory assessment as a distinct institutional practice. Most institutions do not make this investment systematically, because the return is invisible until the capacity is lost.

3. The Reason Retention Principle

The structural analysis suggests a precise criterion for institutional orientation.

Principle II.2 (Reason Retention). *An institution remains oriented only to the extent that it can reconstruct the arguments that originally justified its constraints.*

The word *reconstruct* is doing essential work here and must be distinguished from two weaker capacities that are often confused with it.

The first weaker capacity is *archival retention*: the preservation of documents, records, and stated rules. A sufficiently large archive contains the original justifications in some form. But archival retention does not guarantee that anyone can read those justifications in context, understand the problem they were responding to, or recognize their relevance to current decisions. Archives preserve words. Reconstruction requires understanding.

The second weaker capacity is *recitation*: the ability to reproduce the stated rationale for a rule on demand. Members of an institution may be able to

state why a procedure exists — "this requirement ensures quality control" or "this review process protects against conflict of interest" — without having the interpretive capacity to apply that rationale to novel situations, to recognize when the rationale no longer applies, or to distinguish cases where the original constraint is genuinely relevant from cases where it has become ritual. Recitation is the form without the capacity.

Reconstruction in the full sense requires something stronger: the ability to regenerate the original argument from first principles, to identify the problem the constraint was designed to address, to understand why other approaches were considered inadequate, and to apply that understanding to situations the original designers did not anticipate. An institution that can reconstruct in this sense can determine when a constraint should be preserved, when it should be revised, and when it has outlived its original justification. An institution that can only recite cannot make these determinations reliably.

Observation II.1. *An institution may lose reconstructive capacity while retaining archival retention and recitation. In such a case, constraints continue to be enforced, the stated rationale continues to be reproduced, but the institution has lost the ability to apply the original reasoning to novel circumstances. This is the characteristic state of an institution undergoing silent projection collapse: formally intact, locally coherent, and drifting.*

The reason retention principle has several immediate consequences.

It explains why institutional memory is not reducible to document management. Documents are necessary but not sufficient. What must also be preserved is the interpretive community capable of engaging those documents — the people, practices, and relationships through which the reasoning embedded in constraints is transmitted across time and membership change.

It explains why apprenticeship, mentorship, and extended professional formation persist in domains where orientation matters. The explicit content of legal training, medical education, or scientific apprenticeship can largely be transmitted through documents. What cannot be transmitted through documents alone is the capacity to recognize which constraints apply in which situations, how to weigh competing considerations, and when the formal rule fails to capture the underlying reason. That capacity is transmitted through extended engagement with practitioners who already possess it.

It explains why institutional turnover is more dangerous than it appears. When experienced members leave and are replaced by newcomers whose training was primarily formal, the institution may retain all its rules while losing the interpretive capacity required to apply them correctly. The loss is invisible in the immediate term because the rules continue to be followed. It becomes visible only when novel situations arise that the rules do not clearly cover, at which point the institution must either defer to the rule's surface form or reconstruct a rationale it no longer carries.

It explains why reforms that replace existing constraints with formally superior alternatives sometimes destroy more than they create. If the existing constraint embodied an accumulated understanding that was not fully articulated in its stated rationale, replacing the constraint eliminates that understanding along with the rule. The new constraint may be better by every measurable criterion and still leave the institution less oriented than before, because it has not yet accumulated the interpretive history that would allow it to be applied intelligently.

Institutional Forgetting as an Economic Pressure

Reason decay does not occur by accident. It occurs because the activities required to prevent it are expensive and their returns are invisible until the capacity is lost.

Apprenticeship is expensive. Extended mentorship is expensive. Deliberation is expensive. The maintenance of communities of practice, the writing of case reasoning rather than rule summaries, the study of institutional history including past failures — all of these consume time and attention that could be directed toward measurable outputs. An institution under pressure to increase throughput discovers that the easiest reductions are precisely the orientation-preserving activities that produce no immediate measurable return.

The economic structure of this trade-off is asymmetric. The cost of reducing orientation infrastructure is immediate and measurable: fewer hours in apprenticeship, faster onboarding, more efficient credentialing. The cost of that reduction is deferred and difficult to attribute: the loss of reconstructive capacity that accumulates quietly across years of personnel change. By the time the cost appears, in the form of poor judgment on novel situations or inability to revise constraints intelligently, the institution has long since forgotten the investment decisions that caused it.

The tragedy of institutional forgetting is therefore not that institutions choose to drift. It is that they repeatedly choose local efficiency over orientation capacity without recognizing that those choices are in tension. The institution appears more efficient precisely while becoming less capable of preserving its purpose.

4. Ritualization

Between full reconstruction and complete abandonment, a constraint can enter a third state that is probably more common than either: it can become ritualized.

A ritualized constraint is followed but no longer understood. The rule persists. The behavior persists. What has been lost is the capacity to apply the underlying reasoning to novel situations, to recognize when the rule should be revised, and to distinguish cases where the constraint is genuinely relevant from cases where it is being applied mechanically. The institution continues to enforce the constraint and to produce the stated rationale on demand. But the rationale is recited rather than reconstructed, and the application is formal rather than intelligent.

Observation II.2. *A constraint exists in one of three states: reasoned, when the institution can reconstruct the argument that justifies it; ritualized, when the institution follows the constraint and can recite a rationale but cannot apply the underlying reasoning to novel situations; or abandoned, when the constraint is neither followed nor remembered. Most institutional theory attends to the transition between reasoned and abandoned while neglecting the intermediate state. Ritualization is likely the most common condition of inherited institutional constraints.*

Ritualization is not purely negative. A ritualized constraint preserves behavior even after the understanding behind it has been lost. In stable environments, this may be sufficient. The institution continues to function correctly for the class of situations the original designers anticipated, because the behavior the rule prescribes remains appropriate even when the reason for it is no longer accessible.

The cost of ritualization appears when circumstances change. A novel situation arises that the original rule did not clearly address. An institution with reconstructive capacity can apply the underlying reasoning to the new case, adapt

the rule intelligently, or recognize when the constraint has been made obsolete by changed conditions. An institution that can only recite a rationale cannot do any of these things. It can apply the rule's surface form to the new situation, which may produce absurd results. It can make an exception, which it cannot justify except by appeal to intuition. Or it can refuse to act because the rule does not clearly apply, which may produce paralysis.

Ritualization therefore acts as a slow accumulation of institutional fragility. The institution appears functional in normal conditions and becomes brittle precisely in the situations that most require intelligent judgment. The brittleness is not visible until the novel situation occurs. This is why institutional failure so often appears to originate in exceptional circumstances: the exceptional circumstance did not cause the failure but rather revealed the fragility that ritualization had been accumulating.

The transition from reasoned to ritualized is the most common form of reason decay and the hardest to detect, because it produces no immediate change in institutional behavior. The rule is still followed. The rationale is still stated. Only the capacity for reconstruction has disappeared, and that capacity leaves no visible trace when it goes.

5. Institutional Projection Collapse

The machinery of projection collapse from the previous essay carries over to the institutional setting with one critical modification.

In individual inquiry, projection collapse occurs when the agent's operative goal at each step is a projection of the original goal onto the locally visible context. The projection discards aspects of the original goal that are not visible in the current context. When the context is impoverished, optimization against the projection diverges from the original aim. But in individual inquiry, the original goal remains at least in principle accessible to the agent. The agent can, through deliberate reflection, compare current actions against the original objective and detect divergence.

In institutional projection collapse, this comparison operation itself becomes unreliable. The institution's representation of its original objective is not stored in any single mind. It is distributed across the membership, embedded in practices, encoded imperfectly in documents, and partially carried in the interpretive community whose capacity for reconstruction has been established above. As

that representation degrades — through turnover, through the accumulation of local adaptations that gradually displace original meanings, through the loss of practitioners who carried interpretive capacity informally — the institution's ability to compare its current trajectory against its founding objective diminishes.

Definition II.1 (Institutional Projection Collapse). *An institution undergoes projection collapse when its operative objectives at each decision cycle are projections of the founding objective onto the currently visible institutional context, and when the cumulative displacement between current operative objectives and the founding objective grows across decision cycles even as each individual transition appears locally reasonable. Institutional projection collapse is advanced when the institution has lost sufficient reconstructive capacity that it cannot detect the displacement by internal comparison.*

The institutional version of projection collapse has a feature that the individual version does not: it can become self-sealing.

In individual inquiry, even an agent who has drifted substantially from their original objective retains the cognitive capacity to notice the drift if prompted to compare current activity against the original aim. The original aim was present in the same mind that is now engaged in drifted activity. The comparison is difficult but possible.

In an institution that has lost reconstructive capacity, the comparison is not merely difficult. The institution lacks the representation of the original objective against which comparison would be made. Every internal metric is consistent with the current operative objectives. Every procedure is internally coherent. Every incentive is aligned with the current projection. The institution measures itself against its present configuration and finds that configuration satisfactory. Drift is not detectable from within because the mechanisms required to detect it — access to the original objective, the capacity to reconstruct it, the interpretive community that could recognize divergence — have themselves degraded.

Principle II.3 (Invisibility of Advanced Drift). *Projection collapse is structurally invisible to a sufficiently drifted institution. An institution cannot directly observe the disappearance of an objective it no longer represents.*

This proposition explains why institutional failures so often appear sudden to external observers. The failure did not occur suddenly. It accumulated across many locally reasonable decisions over extended time. But internal metrics showed continuous improvement, or at least acceptable performance, throughout the accumulation. The moment of visible failure is not the moment the institution began to drift. It is the moment external circumstances made the drift impossible to ignore — when the gap between current trajectory and original objective became large enough to produce unmistakable results, or when an external frame of reference made the drift legible for the first time.

The self-sealing character of advanced institutional drift also explains why reform is so difficult. A reform that originates entirely inside a drifted institution is generated from within the drifted context. It will tend to optimize against the current operative objectives rather than against the founding objective. A reform that originates from outside may correctly identify the drift but lacks the interpretive capacity to reconstruct what was lost, and so may replace one projection with another rather than restoring the original orientation.

6. The Hierarchy of Institutional Orientation

Just as orientation in individual inquiry comprises several distinct capacities that scale differently, institutional orientation comprises several distinct functions that are differently susceptible to degradation.

The first is *evaluation*: the capacity to measure performance against explicit criteria. Evaluation is the most tractable level of institutional orientation because it can often be formalized, automated, and distributed. Audits, inspections, rankings, and metrics are all instruments of institutional evaluation. This level of orientation is the most resilient to turnover and the most amenable to scaling. It is also the most susceptible to substituting measurable proxies for genuine objectives — the metric replaces the goal it was designed to approximate.

The second is *goal maintenance*: the active preservation of the institution's operative objectives across time and membership change. Goal maintenance requires more than formal statement of mission. It requires ongoing engagement with the question of whether current activities serve the stated mission, and the capacity to resist local adaptations that gradually displace the mission without formally revising it. Goal maintenance is more fragile than evaluation because it cannot be fully formalized. It depends on the interpretive community that carries the

reconstructive capacity described above.

The third is *trajectory assessment*: the capacity to evaluate whether the institution's current direction, not merely its current state, remains aligned with its objectives. Trajectory assessment requires comparing not only where the institution is but where it is going and whether that direction remains appropriate. This is the level of orientation most commonly missing in institutional practice. Evaluation asks whether current performance is acceptable. Trajectory assessment asks whether the current vector will remain appropriate as circumstances develop. Most institutions conduct evaluation routinely and trajectory assessment almost never.

The fourth is *goal revision*: the capacity to recognize that the founding objective was itself mistaken or incomplete, and to revise it without merely surrendering to drift. Goal revision is the rarest and most difficult form of institutional orientation. It requires the ability to distinguish genuine revision — the principled updating of objectives in response to new understanding — from projection collapse masquerading as revision. Most institutions cannot make this distinction reliably. Constitutional moments, paradigm shifts, and genuine institutional renewals are the exceptional cases where goal revision occurs. They are exceptional precisely because the capacity required for genuine goal revision is so demanding.

Observation II.3. *Most institutions are competent at evaluation, moderate at goal maintenance, poor at trajectory assessment, and almost incapable of genuine goal revision except under the pressure of crisis. This is not a contingent feature of particular institutions. It reflects the relative difficulty of maintaining each level of orientation under normal institutional pressures.*

The hierarchy also explains why the substitution of lower-level for higher-level orientation is so common and so destructive. An institution that has lost trajectory assessment replaces it with more intensive evaluation. More metrics, more audits, more rankings. These can produce locally satisfying results — the metrics improve, the audits pass, the rankings rise — while the institution continues to drift, because the intensified evaluation is conducted against current operative objectives rather than against the founding aim. The institution becomes more rigorously oriented to its current projection and less capable of recognizing the gap between that projection and its original purpose.

7. Repair as Reconstruction

In the previous essay, repair was characterized as the mechanism by which a living system maintains navigability through time. Repair preserves the conditions that keep future states reachable. Applied to individual inquiry, this means preserving structural properties that allow future modification: keeping interfaces clean, maintaining conceptual coherence, preventing commitments from accumulating to the point where they foreclose later revision.

In the institutional setting, repair takes a different form. The structural property that requires preservation is not navigability in a space of actions but the reconstructive capacity that allows the institution to remain oriented to its founding objectives. Institutional repair is the work of maintaining and transmitting the reasons, not merely the rules.

Principle II.4 (Institutional Repair). *Institutional repair is the practice of maintaining reconstructive capacity across time and membership change. It requires preserving not only the constraints an institution operates under but the arguments that made those constraints appropriate, in a form that allows new members to engage them rather than merely inherit them.*

Institutional repair is not the same as institutional maintenance, which primarily concerns the preservation of existing structures. It is also not the same as institutional reform, which primarily concerns the revision of existing structures. Repair is the work that makes both maintenance and reform possible without projection collapse: the ongoing transmission of reasons that allows inherited constraints to be understood, applied intelligently, and revised when appropriate rather than followed ritualistically or abandoned unintelligibly.

The activities through which institutional repair is accomplished are not glamorous and are not easily recognized as productive. They include: extended apprenticeship and mentorship that transmits interpretive capacity rather than only formal knowledge; deliberate engagement with institutional history, including the failures and the reasoning behind rules that exist because something went wrong; the maintenance of communities of practice in which the reasons behind constraints are actively discussed rather than silently assumed; the writing and transmission of case reasoning rather than merely rule statements; and the preservation of dissent and minority views that carry alternative framings of the institution's objectives.

These activities are costly relative to their immediate returns. They consume time and attention that could be directed toward evaluation, production, or expansion. Under pressure to increase search velocity, they are typically the first activities to be reduced. An institution under pressure to act faster will reduce the consultation required before action. It will shorten apprenticeship, compress mentorship, replace case reasoning with rule lookup, and substitute formal credentials for demonstrated capacity. Each of these reductions appears locally rational. The accumulated result is the degradation of the reconstructive capacity on which long-term institutional orientation depends.

Observation II.4. *Compression of apprenticeship, replacement of case reasoning by procedure lookup, and substitution of credentialing for demonstrated interpretive competence frequently appear as efficiency improvements while functioning as mechanisms of reason decay. The institution appears more productive precisely while becoming less capable of preserving its purpose.*

8. Governance as Orientation

Governance is the institutional form of orientation. It is the set of mechanisms by which a collective preserves coherent objectives across time, scale, and membership change — not merely the formal rules by which decisions are made, but the practices through which the reasons behind those rules are maintained and transmitted.

This characterization differs from the standard economic account of governance, which treats it primarily as a coordination mechanism. Coordination is real and important. But governance as coordination is governance at the evaluation level: it concerns whether decisions can be made, implemented, and enforced. Governance as orientation concerns whether the decisions being made remain coherent with the objectives the governance structure was designed to serve.

Definition II.2 (Governance as Orientation). *Governance, in the sense intended here, is the totality of mechanisms through which a collective maintains reconstructive capacity across the four levels of institutional orientation: evaluation, goal maintenance, trajectory assessment, and goal revision. The adequacy of a governance structure is not only a question of whether it can produce decisions, but whether those decisions preserve continuity of aim across time.*

On this characterization, many features of governance that appear irrational from a pure coordination perspective become intelligible as orientation mechanisms. Deliberative delay forces consultation of accumulated context before action. Precedent requires that new decisions be related to prior reasoning rather than generated independently. Constitutional constraint requires that major departures from founding objectives be achieved through processes that make the departure explicit and legible. Separation of powers creates adversarial processes that partially substitute for trajectory assessment by generating external comparisons. Long terms and tenure protect the independence of those who must maintain institutional memory against short-term pressure.

None of these mechanisms are perfectly efficient. All of them impose costs. All of them can be captured or corrupted. But their inefficiency, understood as increased contextual depth, is precisely the feature that makes them function as orientation mechanisms. They slow the institution down at the point where slowing down preserves the capacity for reason reconstruction that would otherwise be lost to search velocity pressure.

The deepest failures of governance are therefore not primarily failures of coordination, incentive alignment, or even competence. They are failures of orientation. An institution can maintain perfect formal structure, complete procedural compliance, and continuous high performance on its current metrics while undergoing advanced projection collapse. The failure appears when external circumstances make the gap between current trajectory and founding objective impossible to ignore. At that point, the institution discovers that its repair capacity has been eroded precisely during the period when its coordination capacity appeared most effective.

Adversarial Orientation

Many of the most durable governance mechanisms preserve orientation not through consensus but through structured disagreement. Peer review, appellate courts, opposition parties, independent inspectorates, and red teams all share a common structure: they create an independent representation of the institution's objective and compare it against current practice. The disagreement is not a failure of coordination. It is the orientation mechanism.

Formally, adversarial processes maintain multiple simultaneous projections of the founding objective G . Where a single projection $\pi(G, C_t)$ may drift without any internal mechanism detecting the divergence, multiple independent

projections $\pi(G, C_t^{(1)}), \pi(G, C_t^{(2)}), \dots$ generated from different contextual positions will disagree as drift advances. The disagreement is itself the signal. An institution whose internal projections remain in perfect agreement may be either genuinely oriented or so thoroughly drifted that all its projections have converged on the same displaced objective. Adversarial processes make that ambiguity legible.

This observation reframes a common institutional tension. The cost of adversarial processes — delay, conflict, the need to justify decisions to critical interlocutors, the possibility of being overturned — is regularly cited as evidence that such processes are inefficient. The reframing is that these costs are orientation infrastructure. They are what makes the independent projection possible. An institution that eliminates adversarial processes to increase decision velocity is trading orientation capacity for search velocity, often without recognizing the exchange.

9. Formal Interpretation

The projection collapse model from the previous essay applies directly to the institutional setting with one structural extension.

Let G denote the founding objective of an institution. At each decision cycle t , the institution's operative objective is a projection $\pi(G, C_t)$ of G onto the institutional context C_t . As before, projection is lossy: $\pi(G, C_t)$ preserves aspects of G visible in C_t and discards the rest.

The institutional context C_t differs from the individual context in a crucial respect. It is not determined solely by the current state of affairs but by the composition, memory, and interpretive capacity of the institution's membership at time t . Formally:

$$C_t = C_t^{\text{state}} \cup C_t^{\text{memory}} \cup C_t^{\text{capacity}}$$

where C_t^{state} captures currently visible institutional conditions, C_t^{memory} captures accessible recorded history and archived reasons, and C_t^{capacity} captures the reconstructive capacity of the current membership — the ability to engage, interpret, and apply the recorded history to novel situations.

Definition II.3 (Reason Decay). *An institution undergoes reason decay at rate $r > 0$*

when C_t^{capacity} decreases over time independently of changes in C_t^{state} and C_t^{memory} . Reason decay occurs when the institution loses practitioners who carry interpretive capacity, when apprenticeship is compressed, or when formal credentials substitute for demonstrated reconstructive ability.

Reason decay is the institutional analog of context impoverishment in individual projection collapse. As C_t^{capacity} decreases, the projection $\pi(G, C_t)$ becomes less faithful even if C_t^{memory} remains rich. The institution may retain every relevant document and still be unable to apply the reasoning those documents embody.

Proposition II.1 (Reason Decay Dominance). *For fixed memory retention, sufficient decline in reconstructive capacity produces projection collapse even when institutional archives remain intact. Formally: there exist institutions for which C_t^{memory} remains constant while $C_t^{\text{capacity}} \rightarrow 0$, such that the projection $\pi(G, C_t)$ diverges from G and the divergence is undetectable by internal comparison.*

This proposition formalizes the essay’s central empirical claim: archives alone do not preserve orientation. A complete and well-maintained archive in the hands of a membership that has lost reconstructive capacity is orientation infrastructure without orientation. The documents exist. The capacity to make them do orienting work does not.

The Invisibility Proposition follows formally from this structure. Detection of projection collapse requires a comparison between G and $\pi(G, C_t)$. This comparison is possible only if the institution maintains a representation of G with sufficient fidelity to serve as a reference. A representation of G is maintained with sufficient fidelity only if C_t^{capacity} is rich enough to reconstruct the original objective from available memory. When reason decay has sufficiently impoverished C_t^{capacity} , the institution cannot maintain a faithful representation of G , the comparison operation fails, and drift becomes internally undetectable.

Conclusion

This essay has argued that institutional drift is structurally inevitable rather than contingent on bad agents or misaligned incentives, because constraints persist more readily than the reasons that justified them. The Reason Retention Principle names the condition for institutional orientation: an institution remains

oriented only to the extent that it can reconstruct the arguments that originally justified its constraints. Reconstruction, not mere archival preservation, is the operative standard.

From this principle, several consequences follow. Institutional projection collapse occurs when accumulated drift displaces operative objectives from founding objectives across many locally reasonable decisions. Advanced projection collapse becomes self-sealing when the institution has lost the reconstructive capacity required to detect the drift. Governance is not primarily a coordination mechanism but an orientation mechanism: the totality of practices through which a collective maintains reconstructive capacity across time and membership change. Institutional repair is not maintenance or reform but the ongoing transmission of reasons that makes both possible without collapse.

The four levels of institutional orientation — evaluation, goal maintenance, trajectory assessment, and goal revision — scale differently. Evaluation is the most tractable and the most susceptible to metric substitution. Goal revision is the rarest and requires the greatest reconstructive capacity. The characteristic failure mode is not incompetence at any level but the substitution of lower-level for higher-level orientation: intensified evaluation replacing trajectory assessment, producing local metric improvement and continued directional drift.

The central question for any institution is therefore not whether it is performing well by its current metrics. It is whether it can still reconstruct the reasons for which those metrics were chosen. An institution that can answer that question is oriented. An institution that cannot is drifting, whether or not any internal measure reflects the fact.

This essay is the second in a pair. The first examined continuity of aim within an individual mind navigating an expanding possibility space. The present essay examines continuity of aim within a collective navigating historical time. In both cases, the central problem emerges when the production of possibilities grows faster than the capacity to preserve orientation. Search expands the space of available actions. Orientation preserves the reason actions are chosen. The scarce resource, in an age of abundant generation, is not production but continuity. Minds require it across iterations. Institutions require it across generations. The medium differs; the structure is the same.

Orientation and Knowledge

On the Fragility of Understanding at Civilizational Scale

Abstract

The previous two essays established that production outpaces orientation at the scale of individual minds and at the scale of institutions. This essay extends the argument to the scale of knowledge traditions. Knowledge accumulates monotonically: results are added to the literature and rarely removed. Understanding does not accumulate in the same way. Understanding is the capacity to construct coherent structure across knowledge — to recognize which results bear on which questions, how findings from different domains constrain each other, and what a new result means in the context of what is already known. That capacity cannot be stored or transmitted by copying; it must be reconstructed by each new understander, at increasing cost as the body of knowledge grows. When knowledge production exceeds the rate at which coherent structure is constructed, understanding debt accumulates. Understanding debt is not merely a backlog of unperformed synthesis. It compounds: subsequent knowledge is produced in the absence of integrations that would have shaped it, deepening the fragmentation it must eventually overcome. The essay introduces the Trajectory Preservation Principle, which unifies the three debts of the series — navigability debt, reason decay, and understanding debt — as distinct manifestations of a single process: the accumulation of states whose generating trajectories can no longer be reconstructed. The literature stores survivors. Understanding stores the selection history. When the selection history disappears, the survivors lose the context that made them significant.

The Architecture of the Series

Before developing the third essay's argument, it is useful to make the architecture of the series explicit. Each essay has identified a productive process, an orientation mechanism, and a failure mode at a different scale of organization.

Scale	Production	Orientation	Failure Mode
Individual	Search	Navigation and repair	Navigability debt
Institution	Constraint accumulation	Reason reconstruction	Reason decay
Knowledge tradition	Knowledge production	Synthesis	Understanding debt

At each scale, production scales more readily than orientation. At each scale, the failure mode is invisible to the standard metrics of the productive process. At each scale, the failure compounds: each unit of unoriented production makes subsequent orientation more expensive. This essay develops the third row of the table and identifies the principle that unifies all three.

1. The Accumulation Problem

Knowledge accumulates monotonically. A result that has been validated enters the literature and is rarely removed. Findings persist. Datasets persist. Proofs persist. Techniques persist. The corpus of any mature discipline is vastly larger than any individual can read, and it continues to grow. By the standard measures of intellectual progress — papers published, citations accrued, discoveries announced, instruments built — the trajectory is continuous and accelerating.

Understanding does not accumulate in the same way. Understanding is relational: it consists not in the possession of individual results but in the construction of coherent structure across results — the recognition of connections, the identification of tensions, the integration of findings from different domains into a picture that is more than the sum of its parts. That construction requires someone to do it, and doing it does not scale with the literature it must integrate. As the literature grows, the work of integration grows at least proportionally. As the literature fragments into increasingly specialized subfields, the work of integration grows faster than the literature itself, because each act of synthesis must now bridge not only the quantity of results but the distance between the conceptual vocabularies in which those results are expressed.

The opening claim of this essay is that there is no conservation law for understanding. Knowledge can be produced faster than understanding can be built. When that happens, the ratio of knowledge to understanding increases without limit. A tradition in that condition has more validated findings than it has the capacity to integrate, and the gap is invisible because the knowledge is real — it is not wrong, it is not lost, it is simply unintegrated. The tradition appears healthy by every standard measure while losing the coherent structure that made accumulated results part of a common inquiry.

2. Knowledge and Understanding

The distinction between knowledge and understanding is the essay's load-bearing joint. It must be made precise.

Knowledge is a validated proposition, result, or finding that has passed the relevant institutional filters: peer review, formal proof, empirical confirmation, replication. Knowledge is propositional. It can be stored, transmitted, and

counted. It is the output of the knowledge-production machinery. A theorem in a textbook is knowledge. A dataset in a repository is knowledge. A technique described in a methods section is knowledge.

Understanding is the construction of coherent structure across knowledge — the capacity to recognize which results bear on which questions, how findings in one domain constrain possibilities in another, where the genuine open problems are, and what a new result would mean in the context of what is already known. Understanding is relational and active. It cannot be stored directly; it must be rebuilt by each mind that possesses it. It is not the output of any machine.

The asymmetry in how these two things are transmitted is the source of the problem this essay addresses. Knowledge transmits mechanically. A paper can be copied, a proof transcribed, a dataset downloaded, at effectively zero marginal cost per recipient. Understanding transmits only through reconstruction. The recipient must not merely receive the proposition but rebuild the web of connections that makes it significant — understand why this result and not another, what it rules out, how it relates to the questions that motivated the inquiry, what it changes about the landscape of what is now possible. That reconstruction cannot be completed by copying. It requires engagement with a body of context that itself requires reconstruction to engage.

Principle III.1 (Knowledge–Understanding Asymmetry). *Knowledge transmits mechanically. Understanding transmits only through reconstruction. As knowledge production accelerates, the reconstruction burden on each new understander increases, while the time and cognitive resources available for reconstruction do not increase proportionally.*

The asset-maintenance analogy makes the structural consequence clear. Knowledge behaves like a capital asset: it accumulates, persists, and can be transferred at low marginal cost. Understanding behaves like maintenance: it must be continuously performed, cannot be stored, and the scale of the maintenance obligation grows with the asset base it must service. A tradition that treats understanding as optional performs the same error as a firm that treats maintenance as optional: the asset base grows while the capacity to use it degrades. Crucially, the error is structurally incentivized. Assets appear on the ledger as growth. Maintenance appears as cost. Any system optimizing visible returns will therefore systematically underproduce understanding relative to knowledge

— not through negligence or bad values, but through the ordinary operation of incentives that reward the measurable and discount the deferred.

A second and subtler asymmetry concerns what is transmitted alongside the result. A scientific result is not merely a proposition. It is the residue of a process: of failures, corrections, anomalies, alternative theories that did not survive, repairs that were made when predictions failed, and revisions that converged on the formulation that eventually appeared in print. The literature records the survivor. It does not systematically record the selection process that produced the survivor. Understanding includes access to that selection history: knowing not only what the result is but what it survived, what it ruled out, and what would have to be true for it to be revised. Without the selection history, the result can be reproduced but not extended, applied but not adapted, cited but not built upon in the deepest sense.

Principle III.2 (Selection History). *The literature stores survivors. Understanding stores the selection history — the record of alternatives considered and rejected, failures that shaped the surviving formulation, and criteria by which competing approaches were evaluated. Selection history cannot be reconstructed from results alone. Its loss is therefore irreversible in a way that the loss of a result is not.*

This observation motivates a distinction that will carry the essay's formal apparatus. There are two types of trajectory whose reconstructability matters for orientation, and they are not equivalent.

A *causal trajectory* is the sequence of states and decisions that produced the current configuration: how things came to be as they are, what path was taken through possibility space, what sequence of choices converged on the surviving formulation. Causal trajectories can in principle be reconstructed from sufficiently rich archives, because the path taken leaves traces in the record.

A *selection trajectory* is the set of alternatives that were considered and rejected, and the criteria by which rejection occurred. Selection trajectories are structurally harder to preserve than causal trajectories, because rejected alternatives leave weaker traces than the surviving path. A validated result records that this formulation survived. It does not record what it survived against, what anomalies ruled out the alternatives, or what would have to be true for the surviving formulation to be overturned. The selection trajectory is what understanding preserves that knowledge alone cannot recover, and its loss is the deeper of the

two losses.

This distinction clarifies the structure of all three debts in the series. Navigability debt is primarily causal trajectory loss: accumulated structural commitments obscure the paths that led to the current state. Reason decay involves both: the causal trajectory of a constraint may survive in institutional archives, but the selection trajectory — what the constraint was chosen against, what problems the rejected alternatives failed to solve — decays first and fastest. Understanding debt is primarily selection trajectory loss at civilizational scale: results accumulate while the failure landscape that made each result significant, and the alternatives it ruled out, disappear.

3. Understanding Debt

When knowledge production exceeds the rate at which coherent structure is constructed across knowledge, a liability accumulates. This essay calls that liability understanding debt.

The term is chosen to emphasize the dynamic rather than the static dimension of the problem. A gap implies a fixed distance between two things. Debt implies accumulation over time, compounding, and increasing future cost. Understanding debt is not merely the observation that synthesis is incomplete at any given moment. It is the claim that unsynthesized knowledge generates an obligation that grows, and that deferring the obligation makes it more expensive to retire.

Principle III.3 (Understanding Debt). *Every increase in accumulated knowledge creates an associated burden of reconstruction and integration. When knowledge production exceeds the rate at which coherent structure is constructed across knowledge, understanding debt accumulates. Understanding debt increases the future cost of synthesis: subsequent knowledge is produced in the absence of integrations that would have shaped it, deepening the fragmentation that synthesis must eventually overcome.*

The compounding mechanism is the crucial addition to a simple backlog model. Technical debt in software provides an illuminating analogy. When structural problems in a codebase are deferred, subsequent development does not wait for the repair. New code is written around the unrepaired structure, adapting to it, depending on it, building abstractions over it. The cost of eventually repairing the original problem therefore increases with time, because repair now requires

disentangling not only the original problem but all the subsequent structure that was built in its absence. The debt does not merely accumulate; it ramifies.

The same mechanism operates in knowledge traditions. When an integration between two bodies of results is deferred, research in each area does not pause. New results are produced within each area, framed by the assumptions and conceptual vocabularies of that area, without benefit of the integration that would have constrained or redirected them. The questions asked, the methods developed, the concepts refined — all of these are shaped by the absence of the integration. When the integration is eventually attempted, it must now accommodate not only the original bodies of results but the subsequent development that occurred in their mutual absence. The cost is not the cost of the original integration but the cost of that integration plus the cost of reconciling everything that grew in the gap.

Observation III.1 (Compounding Understanding Debt). *Understanding debt increases the future cost of synthesis. As debt accumulates, subsequent knowledge production occurs in the absence of integrations that would otherwise have influenced theory construction, educational pathways, and research priorities. The effort required to retire a unit of debt therefore tends to increase with existing debt levels. Understanding debt does not merely accumulate; it shapes what gets produced, deepening the fragmentation that future synthesis must overcome.*

The compounding observation also explains why understanding debt tends to be invisible until it becomes severe. At early stages, the missing integrations are invisible because the research programs that would have been redirected by them appear locally productive. Papers are published. Results are obtained. Careers advance. The absence of synthesis is not registered as absence because the absence has not yet produced visible failures — only invisible redirections of research effort toward questions that would not have been the most important ones had the integration occurred. The cost of compounding understanding debt is therefore primarily a cost in misdirected inquiry: the questions not asked, the connections not recognized, the redundant work performed in parallel by researchers who did not know the other pursuit existed.

4. Fragmentation as Structural Drift

When understanding debt accumulates to sufficient levels, a knowledge tradition undergoes structural fragmentation. Fragmentation is not the division of a field into subfields, which is a normal and often productive organizational response to growth. Fragmentation is the condition in which the subfields of a tradition have lost genuine contact with each other and with the founding questions that originally motivated the tradition as a whole.

In a fragmented field, subfields develop that share terminology but not understanding. Results accumulate in one area that would constrain or transform work in another if anyone carried the synthetic understanding to apply them. Questions are pursued in parallel by researchers who are unaware of the other pursuit. Foundational assumptions that were adopted provisionally, because they were the best available at the time, become invisible and unchallenged because no one has the synthetic overview required to see them as assumptions rather than facts. Methods become detached from the problems that motivated their development and are applied to problems for which they are not suited, because the practitioners who apply them have not reconstructed the selection history that revealed their limitations.

Fragmentation is the third essay's analog of institutional drift. Just as an institution can retain all its formal structure while losing the reasons that made that structure appropriate, a field can retain all its validated results while losing the coherent structure that made those results part of a common inquiry. The field is not wrong. No individual result is incorrect. The failure is at the level of the whole: the field no longer functions as an integrated inquiry but as a collection of local inquiries whose connection to each other and to the original motivating questions has become attenuated.

Observation III.2. *A knowledge tradition undergoes fragmentation when local density within subfields increases while global coherence across subfields decreases. Each subfield becomes more internally consistent and more isolated from the others. The founding questions become visible only from positions that no longer exist in the institutional structure of the discipline.*

The invisibility of fragmentation follows from the same logic as the invisibility of institutional drift. A fragmented field measures itself by local metrics: publication rates, citation counts, grant funding, conference attendance, student

enrollment. These metrics are sensitive to local density — to whether subfields are productive — and insensitive to global coherence — to whether the subfields remain in genuine contact with each other and with the founding questions. A field can show continuous improvement on every standard metric while its global coherence degrades.

The comparison operation that would detect fragmentation — comparing current subfield trajectories against the founding questions — requires a synthetic overview that fragmentation has made rare or absent. The field cannot observe its own fragmentation by internal means, because the mechanisms required to detect it are precisely those that fragmentation has eroded.

5. Knowledge as Asset, Understanding as Maintenance

The structural tendency toward understanding debt deserves its own treatment, because it is not merely the observation that synthesis is difficult. It is the observation that the incentive structure of knowledge production systematically discourages the maintenance of understanding, for reasons that are structural rather than cultural.

Knowledge production yields immediate, visible, and attributable returns. A paper is published with authors' names attached. A discovery is announced and credited. A technique is developed and cited. A dataset is released and downloaded. These outputs can be counted, ranked, and rewarded. They accumulate in the record and remain attributed to their producers indefinitely.

Synthesis yields returns that are diffuse, deferred, and difficult to attribute. A synthetic understanding of how two bodies of results constrain each other may redirect research across an entire subfield, but the redirection is distributed across many researchers and many projects, none of which is likely to trace itself to the synthetic act that enabled it. Synthesis is also slow: a genuine integration of two bodies of knowledge requires mastery of both, which requires the reconstruction burden that the Knowledge–Understanding Asymmetry makes increasingly expensive as both bodies grow. The person who performs the synthesis spends years acquiring the reconstruction capacity to do it, produces few citable outputs during that acquisition, and when the synthesis is complete, produces a result that is difficult to cite because it is structural rather than propositional — it changes the landscape rather than adding a point to it.

Principle III.4 (Maintenance Underinvestment). *Any system that rewards knowledge production by immediate, attributable output will systematically underinvest in synthetic understanding, because synthesis is maintenance rather than production. Maintenance preserves the coherence of the asset base but does not add to it. A tradition that treats synthesis as optional will degrade its understanding capacity at a rate proportional to its knowledge production rate.*

This principle explains why the tendency toward understanding debt is structural rather than contingent. It is not that contemporary knowledge traditions happen to undervalue synthesis. It is that any sufficiently successful knowledge-producing tradition will tend to undervalue synthesis unless it deliberately creates institutions to protect it. The protection requires actively rewarding activities that have no immediate output — understanding as an end in itself, synthesis as a distinct intellectual contribution, reconstruction capacity as a form of expertise that deserves recognition independent of the results it enables. Most knowledge traditions do not maintain this protection systematically, and those that establish it tend to erode it under production pressure.

6. Synthesis as Repair

Across the three essays, repair takes a different form at each scale while serving the same structural function: it is the activity that preserves orientation against the accumulation that threatens it.

In the first essay, repair preserved navigability — the structural conditions that kept future states reachable. A system that is never repaired accumulates commitments that foreclose paths. Repair reopens them.

In the second essay, repair reconstructed reasons — the interpretive capacity that kept constraints intelligible. An institution that is never repaired loses the ability to apply its inherited constraints to novel situations. Repair transmits the selection history of the constraints alongside the constraints themselves.

In the third essay, the analogous activity is synthesis: the construction and transmission of integrative understanding that keeps the accumulated body of knowledge oriented toward genuine questions.

Synthesis is not review. A review article summarizes what is known in a subfield, organized for consumption by specialists in that subfield. Synthesis constructs coherent structure across subfields — it identifies what findings in one area

imply for open questions in another, what tensions between subfields reveal about shared foundational assumptions, what the shape of the whole looks like from a position that no specialist occupies. Synthesis is the activity that makes accumulated knowledge into something more than an archive.

Principle III.5 (Synthetic Repair). *Synthesis is the repair mechanism of knowledge traditions. Its function is not to add to the body of knowledge but to maintain the coherence of the structure across knowledge — to prevent fragmentation from advancing, to transmit selection histories alongside results, and to keep the founding questions visible from within the accumulated literature. The value of synthesis increases as knowledge production accelerates, because understanding debt widens faster than any specialist inquiry can close it.*

The people who perform synthesis are not the most productive by standard metrics. They may produce fewer results per year than high-output specialists. They may spend years acquiring the reconstruction capacity required to synthesize bodies of knowledge that specialists have spent careers producing. But their contribution is not to the quantity of knowledge. It is to the coherence of understanding. When synthesis is absent, the knowledge-to-understanding ratio increases unchecked. When synthesis disappears entirely, the tradition is left with an archive it can no longer navigate — a library without the understanding required to use it.

Synthesis also transmits what pure archival preservation cannot: the selection history. A synthesis that integrates two bodies of results must engage not only the results but the failure landscape that shaped them — what was tried, what did not work, what the surviving results ruled out. In transmitting that engagement, synthesis makes the selection history accessible to a new generation of researchers in a form they could not have reconstructed from the literature alone. This is why genuine pedagogy is a form of synthesis, and why the best teachers are not those who know the most results but those who can reconstruct the selection history that makes the results significant.

7. Formal Interpretation

This section develops the formal apparatus for understanding debt and unifies it with the projection collapse framework from the previous two essays.

Understanding Debt

Let K_t denote the body of validated knowledge at time t , with $|K_t|$ measuring its size. Let $\beta(t)$ denote the rate at which knowledge is integrated into coherent structure — the rate of synthesis. Understanding debt accumulates whenever production exceeds synthesis:

Definition III.1 (Understanding Debt).

$$D_t = \int_0^t \left(\frac{d|K_\tau|}{d\tau} - \beta(\tau) \right) d\tau.$$

D_t is the accumulated stock of unintegrated knowledge at time t . $dD_t/dt > 0$ whenever knowledge production exceeds coherence construction.

The compounding mechanism enters through the cost of synthesis. Let the cost of integrating one unit of unintegrated knowledge increase with existing debt:

$$c(D_t) = c_0(1 + \lambda D_t), \quad \lambda > 0.$$

If available synthesis effort at time t is E_t , the achievable synthesis rate is:

$$\beta(t) = \frac{E_t}{c(D_t)} = \frac{E_t}{c_0(1 + \lambda D_t)}.$$

Substituting into the debt equation:

$$\frac{dD_t}{dt} = \frac{d|K_t|}{dt} - \frac{E_t}{c_0(1 + \lambda D_t)}.$$

As D_t grows, the denominator increases, so fixed synthesis effort retires less debt per unit time. This formalizes the compounding observation: debt reduces the future effectiveness of synthesis, independently of whether knowledge production accelerates.

Proposition III.1 (Threshold for Unbounded Debt). *If knowledge grows exponentially, $|K_t| = K_0 e^{gt}$, but synthesis capacity grows at most linearly, $E_t \leq a + bt$, then there exists a threshold time t^* after which $dD_t/dt > 0$ regardless of synthesis effort.*

Beyond t^ , understanding debt grows without bound unless synthesis capacity scales at least exponentially with knowledge production.*

The compounding factor λD_t in the denominator of $\beta(t)$ reinforces this result: even if synthesis effort were to scale proportionally with knowledge production, the declining efficiency of synthesis effort under accumulated debt would still produce unbounded growth beyond a sufficiently high debt level. The system has no self-correcting mechanism once compounding is sufficiently advanced.

A further consequence concerns where synthesis activity concentrates as debt accumulates.

Proposition III.2 (Synthesis Suppression). *If the cost of synthesis increases with accumulated understanding debt, then beyond a threshold debt level the marginal effectiveness of cross-domain integration declines faster than the marginal effectiveness of within-domain synthesis. Synthesis activity therefore becomes increasingly concentrated in local domains rather than cross-domain integration, accelerating fragmentation precisely where fragmentation is already most damaging.*

The mechanism is the compounding cost function $c(D_t)$. Cross-domain synthesis requires reconstruction of two or more independent bodies of knowledge before integration can begin; its cost therefore scales with the debt in each domain independently. Within-domain synthesis requires reconstruction of only one body; its cost scales with that domain's local debt. As global debt rises, the relative cost of cross-domain work increases faster than local work. Rational allocation of fixed synthesis effort therefore shifts progressively toward local domains. The result is that as understanding debt grows, the synthesis that does occur becomes less capable of arresting fragmentation, because it is concentrated in the subfields already most internally coherent rather than in the cross-domain connections that global coherence requires.

This proposition formalizes the essay's central structural claim. Understanding debt is not a problem that additional resources straightforwardly solve. When knowledge production is exponential and synthesis capacity is bounded by the reconstruction burden each understander must carry, debt accumulation becomes structurally inevitable beyond a threshold. The problem is not a shortage of synthesizers. It is a mismatch in scaling regimes.

Fragmentation

Represent a knowledge tradition as a graph $G_t = (V_t, E_t)$, where vertices are validated results and edges represent understood relations between results. Partition the vertex set into subfields $S_1, \dots, S_n$.

Define local density within subfields:

$$\rho_{\text{local}}(t) = \frac{1}{n} \sum_{i=1}^n \frac{|E_i(t)|}{|V_i(t)|^2},$$

where E_i are edges internal to subfield S_i .

Define global coherence as the density of cross-subfield integration:

$$\kappa(t) = \frac{|E_{\text{cross}}(t)|}{|V_t|^2}.$$

Definition III.2 (Fragmentation). *A knowledge tradition undergoes fragmentation when*

$$\frac{d\rho_{\text{local}}}{dt} > 0 \quad \text{while} \quad \frac{d\kappa}{dt} < 0.$$

Fragmentation is advanced when $\kappa(t) < \kappa_{\min}$, where $\kappa_{\min}$ is the minimum coherence required for cross-subfield reasoning to remain tractable.

Projection Collapse of Founding Questions

Let Q denote the founding questions of a knowledge tradition. Each subfield S_i carries a projection $Q_i = \pi_i(Q)$ of the founding questions onto its locally visible context. As fragmentation advances, the projections diverge:

$$\Delta_Q(t) = \frac{1}{n} \sum_{i=1}^n d(Q_i, Q).$$

Projection collapse of founding questions occurs when $d\Delta_Q/dt > 0$: the average subfield projection moves away from the original questions. Advanced fragmentation obtains when no subfield projection retains sufficient fidelity to recover Q :

$$\forall i, \quad R(Q_i) < \theta,$$

where $R(Q_i)$ measures the reconstructive fidelity of projection Q_i and θ is the minimum threshold for recovery.

The Visibility Failure

Let M_t denote the standard internal metrics of a knowledge tradition: publication rate, citation density, grant volume, enrollment, conference activity. These metrics are sensitive to local production and local density: $M_t = f(|K_t|, \rho_{\text{local}}, \dots)$.

Fragmentation depends on global coherence and accumulated debt: $F_t = g(D_t, \kappa(t), \Delta_Q(t))$.

If standard metrics are insensitive to global coherence and debt — if $\partial M_t / \partial \kappa \approx 0$ and $\partial M_t / \partial D_t \approx 0$ — then:

Proposition III.3 (Invisibility of Fragmentation). *A sufficiently fragmented knowledge tradition cannot observe its own fragmentation by standard internal measures. The metrics by which fields assess their health are insensitive to the quantities that determine whether understanding is being preserved.*

Trajectory Preservation

The distinction between causal and selection trajectories, introduced in Section 2, maps cleanly onto the formal apparatus. The reconstructive fidelity $R(Q_i)$ of a subfield projection measures how much of the founding question Q can be recovered from Q_i . Causal fidelity — recovery of the path taken — is relatively tractable from archives. Selection fidelity — recovery of the alternatives that failed and the criteria by which they were rejected — is not recoverable from results alone. Understanding debt is therefore primarily an accumulation of selection trajectory loss: the E_{cross} edges in the knowledge graph encode understood relations between results, but they do not encode why the alternative relations were ruled out.

Principle III.6 (Trajectory Preservation). *A system remains oriented only to the extent that future agents can reconstruct both the causal trajectories that generated its present state and the selection trajectories — the alternatives considered and rejected — that made those causal paths meaningful. Orientation failure occurs when state accumulation exceeds the reconstructability of both trajectory types.*

Reconstruction reachability — the capacity of future agents to recover both trajectory types — is therefore the quantity that all three forms of orientation debt erode. Navigability debt closes paths forward. Reason decay closes paths back through the institutional past. Understanding debt closes paths back

through the intellectual past. In each case, the productive process generates states. The orientation process maintains access to the trajectories that made those states the ones that survived. When states accumulate faster than trajectory reconstructability is maintained, the system loses orientation not all at once but gradually, invisibly, and in a manner that compounds.

Conclusion

This essay has argued that knowledge traditions face a structural analog of the orientation failures identified at individual and institutional scales. The Knowledge–Understanding Asymmetry establishes that understanding transmits only through reconstruction, while knowledge transmits mechanically. This asymmetry makes understanding debt structurally inevitable under knowledge production pressure: maintenance is systematically underinvested relative to production because maintenance produces no immediately attributable output. Understanding debt compounds because deferred integration shapes subsequent production, deepening the fragmentation that synthesis must eventually overcome. Fragmentation is the advanced form of the failure: local subfield density increases while global coherence decreases, and the founding questions become invisible from within any specialist position.

Synthesis is the repair mechanism of knowledge traditions. It functions not to add results but to maintain the coherence of structure across results — to preserve selection histories alongside survivors, to keep founding questions visible from within the accumulated literature, and to prevent fragmentation from becoming self-sealing. The value of synthesis increases as knowledge production accelerates, for the same reason that repair becomes more important as system generation increases and institutional reconstruction becomes more important as constraint accumulation accelerates. In every case, orientation is the activity that keeps the productive process from outrunning itself.

The three essays in this series have examined orientation failure at three scales.

The first examined the individual mind navigating an expanding possibility space. Search scales; orientation does not. The failure is navigability debt: the accumulation of structural commitments that foreclose future paths. The repair is navigation — the activity through which the agent maintains access to the future.

The second examined the institution navigating historical time. Constraints accumulate; reasons decay. The failure is institutional drift: the gradual displacement of founding objectives by locally visible proxies. The repair is reconstruction — the transmission of reasons alongside rules, keeping inherited constraints intelligible across generations.

The third has examined the knowledge tradition navigating civilizational accumulation. Knowledge accumulates; understanding does not scale proportionally. The failure is fragmentation: the loss of coherent structure across a body of results that individually remain valid. The repair is synthesis — the construction of integrative understanding that keeps accumulated knowledge oriented toward the questions that motivated its production.

At each scale, the failure mode is invisible to the metrics of the productive process. At each scale, the failure compounds. At each scale, the repair activity appears as inefficiency from within the perspective of the productive process. And at each scale, the common underlying structure is the same: states accumulate faster than the trajectories that generated them can be reconstructed. Navigability debt, reason decay, and understanding debt are three names for the same structural process operating at different scales and through different media.

A simpler formulation of the same invariant runs through all three essays. Every debt in the trilogy arises because systems preserve nouns more effectively than verbs. Search results more effectively than navigation. Constraints more effectively than reasons. Knowledge more effectively than understanding. States more effectively than trajectories. In each case, the residue of the process is preserved while the process that produced the residue is lost. Civilization becomes disoriented when it preserves the products of processes more effectively than the processes themselves. This is not a cultural failing or a failure of values. It is structural: products are propositional and can be copied; processes are active and must be reconstructed. The asymmetry in transmission cost between the two is the common root of all three forms of orientation debt.

Principle III.7 (Trajectory Preservation). *A system remains oriented only to the extent that future agents can reconstruct the trajectories — both causal and selective — that generated its present state. Orientation failure occurs when state accumulation exceeds trajectory reconstructability. Navigability debt, reason decay, and understanding debt are distinct manifestations of this single process at different scales of organization.*

What accelerates is always visible. The literature grows. The institution expands. The archive fills. What preserves orientation is always invisible until it is gone: the paths not foreclosed, the reasons not forgotten, the selection histories not lost. The measure of a tradition's health is therefore not what it has accumulated but what it can still reconstruct. The literature stores survivors. Understanding stores the selection history. When the selection history disappears, the survivors lose the context that made them significant — and the tradition, however large its archive, has lost its orientation.

Epilogue: Orientation as Reconstruction

These three essays have arrived, by different routes, at the same place.

The first began with the observation that search capacity has expanded faster than orientation capacity, and derived from that asymmetry a set of consequences about how individual inquiry should be structured when generation is abundant. The second began with the observation that institutional constraints outlive the reasons that justified them, and derived from that asymmetry a theory of institutional drift and repair. The third began with the observation that knowledge accumulates faster than understanding can be constructed, and derived from that asymmetry a formal account of fragmentation and understanding debt.

By the end of the third essay, the deeper invariant had become visible: in every case, products accumulate more readily than the processes that generated them. This is the Artifact–Process Asymmetry, and it is the common root of all three debts.

Principle III.8 (Artifact–Process Asymmetry). *The products of a process are generally easier to store, transmit, and measure than the process that produced them. Orientation failure occurs when artifact preservation systematically exceeds process reconstructability. Search results, institutional constraints, and validated knowledge findings are all artifacts in this sense. Navigation, reason, and understanding are all processes. The trilogy’s three debts arise from the same structural tendency: artifacts are preserved while processes decay.*

The recurring structure can be stated compactly. In each essay a production process generates artifacts. An orientation process maintains access to the generative trajectories behind those artifacts. When artifact accumulation outpaces trajectory reconstructability, orientation fails — invisibly at first, then in ways that cannot be addressed from within the system, because the resources required to detect and repair the failure are themselves products of the orientation process that has degraded.

Scale	Artifact	Generative Process	Trajectory Lost
Individual	Search result	Navigation	Goal continuity
Institution	Constraint	Reason	Selection history of rules
Knowledge tradition	Result	Repair history	Selection history of findings

The Trajectory Preservation Principle, stated in Part III, can now be read as the formal expression of the Artifact–Process Asymmetry:

A system remains oriented only to the extent that future agents can reconstruct the trajectories — both causal and selective — that generated its present state. Orientation failure occurs when state accumulation exceeds trajectory reconstructability.

This principle is not new to this trilogy. It has appeared, in various vocabularies, across a wider body of work. The present epilogue places the trilogy in relation to that work.

Connections to Prior Frameworks

CLIO and projection loss. The Constraint, Locality, and Irreversible Ontology framework [3] treats projection as the fundamental operation by which a representation maps onto a local context. Projection is necessarily lossy: a projection $\pi(G, C_t)$ preserves the aspects of a goal G that are visible in context C_t and discards the rest. Repeated projection without repair produces cumulative displacement — projection collapse — in which the operative goal diverges from the original while appearing locally coherent.

The trilogy’s three debts are all instances of projection collapse operating at different scales and through different media. Navigability debt is projection collapse of goal continuity across search iterations. Reason decay is projection collapse of institutional objectives across generational transitions. Understanding debt is projection collapse of founding questions across the fragmentation of a knowledge tradition. CLIO provides the common formal language: in every case, repeated projection onto impoverished context produces drift that is invisible from within the projection sequence and detectable only from an external reference that the drift may itself have made inaccessible.

RSVP and reachability volume. The Relativistic Scalar-Vector Plenum framework [2] is concerned with the geometry of reachable states: the question is not whether a state can be produced but whether it can be reached without foreclosing future movement. A central quantity in RSVP is the reachability volume $\text{Vol}(\mathcal{A})$ — the measure of admissible future states accessible from the current position.

Each of the trilogy's debts corresponds to a reduction of reachability volume at a different scale. Navigability debt in individual inquiry reduces the set of future paths accessible from the current system state. Reason decay in institutions reduces the set of future adaptive responses available to an institution confronting novel situations. Understanding debt in knowledge traditions reduces the set of productive research trajectories accessible from the current body of knowledge. In each case, the productive process appears to increase capability — more results, more constraints, more knowledge — while the orientation failure reduces future reachability. What is accumulated is not capability but closure.

Repair Theory and the primacy of process. The repair-centered view of knowledge, developed across several prior monographs [5], treats states as residues of repair histories rather than primary objects. A theorem is not merely true; it is what survived a particular sequence of challenges, corrections, and refinements. A constraint is not merely a rule; it is the residue of the reasoning that addressed a particular class of failures. Understanding is access to the repair history — the record of what was tried, what failed, and why the surviving formulation was preferred over the alternatives.

Repair Theory provides the deepest connection to the Artifact–Process Asymmetry. If states are residues of processes, and if processes cannot be recovered from residues alone, then any system that preserves states more effectively than processes is systematically destroying the interpretive capacity required to use what it has preserved. This is precisely the structure of all three debts. The trilogy's contribution is to show that this structure appears at individual, institutional, and civilizational scales — that Repair Theory is not a theory about one domain but a general principle about how orientation fails whenever artifact accumulation outpaces process reconstructability.

Admissibility and distinction preservation. The Admissibility Field, developed in connection with the HYDRA framework [4], treats admissibility as the preservation of distinctions necessary for future reconstruction. A transforma-

tion is admissible if it does not eliminate distinctions that a future agent would need to reconstruct the trajectory that produced the current state. Inadmissible transformations close off possibilities not by making them unreachable in principle but by eliminating the distinctions required to identify and pursue them.

The trilogy's three orientation mechanisms — navigation, reason reconstruction, and synthesis — are all admissibility-preserving activities. Navigation preserves the distinctions between productive and unproductive future paths. Reason reconstruction preserves the distinctions between constraints that are appropriate in a given context and constraints that have become ritualized or obsolete. Synthesis preserves the distinctions between results that bear on each other and results that are merely co-temporal. When these activities are compressed or eliminated under production pressure, the corresponding distinctions collapse. The system retains more artifacts while becoming less capable of distinguishing among them in ways that matter.

Reconstructive memory. The account of memory offered in the cognitive literature by Bartlett [22] and developed by subsequent researchers [23] treats remembering not as retrieval from storage but as active reconstruction. A memory is not a record that is played back; it is a process of re-entering a trajectory, of reassembling context sufficient to make the past event coherent in the present. Memory fails not when the record is deleted but when the reconstructive capacity to re-enter the trajectory degrades.

This account maps directly onto the trilogy's account of orientation. Navigability is memory of future paths. Reason retention is institutional memory of justifications. Understanding is disciplinary memory of selection histories. In each case, the relevant memory is not archival but reconstructive. An institution that has lost its reasons has not lost its records; it has lost the capacity to reconstruct the trajectory through which those records were generated. A knowledge tradition that has accumulated understanding debt has not lost its results; it has lost the capacity to reconstruct the processes through which those results became significant.

A Generalized Orientation Deficit

Each essay introduced a quantity that accumulates when orientation fails. Navigability debt N_t accumulates when structural commitments foreclose future

paths. Reason decay R_t accumulates when institutional constraints are transmitted without the reasons that justified them. Understanding debt D_t accumulates when knowledge is produced faster than coherent structure can be constructed across it.

These three quantities are formally analogous. Each is an integral of a production rate minus an orientation rate over time. Each grows when production exceeds orientation. Each compounds: accumulated deficit increases the cost of subsequent orientation, which increases the rate at which deficit accumulates. Each is invisible to the standard metrics of the productive process at its scale.

A generalized orientation deficit can be defined as the vector:

$$\Omega_t = (N_t, R_t, D_t),$$

whose components measure reconstruction failure at individual, institutional, and civilizational scales respectively. A fully oriented system has $\Omega_t \approx \mathbf{0}$: navigability is maintained, reasons are reconstructible, and understanding tracks knowledge production. Orientation failure appears as growth in one or more components of Ω_t , typically in a sequence: individual navigability fails first, institutional reason decay follows as individuals and teams lose coherent aim, and civilizational understanding debt accumulates last and most slowly as the knowledge traditions that might have corrected the prior failures have themselves fragmented.

The connection to RSVP-style reachability manifolds is direct. Each component of Ω_t corresponds to a reduction in admissible volume at its scale. A future monograph might define the orientation manifold as the space of system states for which Ω_t remains below a threshold across all three components, and characterize the dynamics by which productive processes push states toward the boundary of that manifold while orientation processes maintain states within it.

What the Trilogy Does Not Cover

Several natural extensions lie beyond the scope of these essays.

The trilogy treats the three scales as distinct. In practice they interact. Individual navigability failures aggregate into institutional drift when enough individuals

within an institution lose orientation simultaneously. Institutional drift contributes to civilizational understanding debt when the institutions responsible for transmitting and evaluating knowledge have themselves lost the reasons behind their own practices. A more complete account would model these interactions and the conditions under which orientation failure at one scale accelerates failure at others.

The trilogy treats orientation failure as a pathology to be avoided. It does not address the question of when constraint should be imposed — when it is appropriate to formalize, to commit, to stop exploring. The admissibility framework provides one approach to this question: commitment is appropriate when the distinctions required for future reconstruction can be preserved within the committed structure. But the exact conditions for appropriate commitment remain underdeveloped here.

The trilogy does not address the question of recovery. Each essay describes mechanisms of repair within a scale — navigation, reconstruction, synthesis — but does not characterize the conditions under which recovery from advanced orientation failure is possible. The self-sealing character of advanced institutional drift and advanced fragmentation suggests that recovery may require external reference frames, constitutional moments, or paradigm shifts that cannot originate from within the drifted system. A theory of orientation recovery would be a natural sequel.

The Deepest Invariant

The trilogy began with a practical observation about search capacity and ended with a structural theorem about trajectory reconstructability. That movement is, in itself, an instance of what the trilogy describes.

The practical observation — that recent systems have made generation abundant — is an artifact: a result that can be stated, transmitted, and measured. The structural theorem — that orientation failure is the loss of trajectory reconstructability — is the generative process behind the artifact: the understanding that makes the observation significant, that connects it to institutional theory and epistemology and the philosophy of science, that allows it to be applied to novel cases the original observation did not contemplate.

The trilogy was an attempt to preserve not only the observation but the selection history: the reasoning that ruled out simpler formulations, the connections

that revealed the observation as a special case of something more general, the process by which a practical concern about search abundance became a structural principle about orientation at every scale.

Whether that attempt has succeeded is a question for reconstruction.

References

- [1] R. Jena, P. Chaudhari, and J. C. Gee. Adaptive Riemannian optimization for multi-scale diffeomorphic matching. *Nature Communications*, 17:4774, 2026. <https://doi.org/10.1038/s41467-026-72508-3>
- [2] Flyxion. Axioms for a falling universe: Field equations of the RSVP framework. *Independent Monograph*, 2024.
- [3] Flyxion. CLIO: Constraint, locality, and irreversible ontology. *Independent Monograph*, 2025.
- [4] Flyxion. HYDRA: Hierarchical dynamics and reachability architecture. *Independent Monograph*, 2025.
- [5] Flyxion. The geometry of closure: Six-stage ontological enlargement. *Independent Monograph*, 2025.
- [6] C. Chenoweth and A. Chenoweth. MEM|8: A wave-based cognitive architecture for multimodal memory integration and consciousness simulation. *Zenodo*, 2025. <https://doi.org/10.5281/zenodo.16436298>
- [7] H. A. Simon. A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.
- [8] H. A. Simon. *The Sciences of the Artificial*. MIT Press, Cambridge, MA, 1969.
- [9] J. G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991.
- [10] D. Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.
- [11] G. Gigerenzer, P. M. Todd, and the ABC Research Group. *Simple Heuristics That Make Us Smart*. Oxford University Press, New York, 1999.
- [12] W. B. Arthur. *Increasing Returns and Path Dependence in the Economy*. University of Michigan Press, Ann Arbor, 1994.
- [13] M. Polanyi. *The Tacit Dimension*. Doubleday, Garden City, NY, 1966.
- [14] D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, Cambridge, 1990.
- [15] E. Ostrom. *Governing the Commons: The Evolution of Institutions for Collective*

- Action*. Cambridge University Press, Cambridge, 1990.
- [16] B. Levitt and J. G. March. Organizational learning. *Annual Review of Sociology*, 14:319–340, 1988.
 - [17] J. G. March and J. P. Olsen. *Rediscovering Institutions: The Organizational Basis of Politics*. Free Press, New York, 1989.
 - [18] T. S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 1962.
 - [19] D. J. de Solla Price. *Little Science, Big Science*. Columbia University Press, New York, 1963.
 - [20] J. T. Klein. *Interdisciplinarity: History, Theory, and Practice*. Wayne State University Press, Detroit, 1990.
 - [21] G. Ryle. *The Concept of Mind*. Hutchinson, London, 1949.
 - [22] F. C. Bartlett. *Remembering: A Study in Experimental and Social Psychology*. Cambridge University Press, Cambridge, 1932.
 - [23] E. Hutchins. *Cognition in the Wild*. MIT Press, Cambridge, MA, 1995.
 - [24] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973.
 - [25] M. E. J. Newman. *Networks: An Introduction*. Oxford University Press, Oxford, 2010.
 - [26] A.-L. Barabási. *Network Science*. Cambridge University Press, Cambridge, 2016.